



Technical Report – Infrared Emitter

T3 - Specialist Technician in Automation,
Robotics, and Industrial Control

Bruna Ferreira Gomes Madeira



Operation an frequency control of 555

The LM555 timer is configured in Astable operating mode. In this configuration, the timing capacitor (C1) repeatedly charges and discharges through resistors R_A and R_B . This continuous charge–discharge cycle forces the output at pin 3 to switch periodically between High and Low logic levels, generating a continuous of square-wave pulses. These pulses are applied to the base of the BD136 transistor, which in turn drives the infrared LED (LD271). The resulting output frequency is determined by the standard astable 555 formula :

Key formulas for a 555 timer in Astable mode:

$$f = \frac{1}{0,693(R_A + 2R_B)C}$$

$$Duty_{OUT} = \frac{R_B}{R_A + 2R_B}$$

From schematic:

$$R_1 = 15k\Omega$$

$$R_2 = 2k7\Omega$$

$$C_1 = 15 \times 10^{-9}F$$

$$P_1 = 10k\Omega$$

Average operating frequency:

$$f_{average} \rightarrow \left(\frac{P_1}{2}\right)$$

$$f = \frac{1}{0,693(R_A + 2R_B)C_1}$$

$$f_{average} = \frac{1}{0,693((15000 + 5000) + (2 \times 2700))15 \times 10^{-9}}$$



$$f_{average} = 3,78kHz$$

Average duty cycle:

$$Duty_{average} = \frac{R_B}{R_A + 2R_B}$$

$$Duty_{average} = \frac{2700}{((15000 + 5000) + (2 \times 2700))}$$

$$Duty_{average} = 0,1063 \times 100$$

$$Duty_{average} = 10,63\%$$

Components determining frequency and min/max range:

The output frequency of the Astable 555 is determined by the timing network formed by R_A ($R_1 + P_1$), R_B (R_2), and capacitor C_1 . The potentiometer P_1 adjusts R_A , modifying the charging time of C_1 . Because the capacitor discharges only through R_B , variations in P_1 influence the charging interval and therefore affect both the oscillation frequency and duty cycle, but do not change the discharge time, which depends solely on R_B .

From schematic:

$$R_1 = 15k\Omega$$

$$R_2 = 2k7\Omega$$

$$C_1 = 15 \times 10^{-9}F$$

$$P_1 = 0 - 10k\Omega$$

Maximum operating frequency:

$$f_{max} \rightarrow P_1 = 0k\Omega$$

$$f = \frac{1}{0,693(R_A + 2R_B)C_1}$$



$$f_{max} = \frac{1}{0,693((15000) + (2 \times 2700))15 \times 10^{-9}}$$

$$f_{max} = 4,71 kHz$$

Minimum operating frequency:

$$f_{min} \rightarrow P_1 = 10 k\Omega$$

$$f = \frac{1}{0,693(R_A + 2R_B)C_1}$$

$$f_{min} = \frac{1}{0,693((15000 + 10000) + (2 \times 2700))15 \times 10^{-9}}$$

$$f_{min} = 3,16 kHz$$

Infrared LED Current Behavior

When the 555 output goes HIGH, the PNP transistor (BD136) turns OFF, and no current flows through the infrared LEDs, resulting in an LED current of 0A. Conversely, when the 555 output goes LOW, the transistor saturates and allows current to flow, turning the LEDs ON. During this period, the LEDs receive the peak current (I_{peak}), which represents the maximum instantaneous current through the LEDs and can be calculated as:

$$I_{peak} = \frac{V_{cc} - V_F - V_{ce(sat)}}{R_4}$$

Since the 555 output is a square wave with a limited duty cycle, the average current (I_{avg}) over one complete cycle is lower than the peak current and is determined by multiplying the peak current by the duty cycle:

$$I_{avg} = I_{peak} \times Duty$$



In this circuit, a duty cycle of 13% is used to limit power consumption. This approach ensures that the LEDs can safely handle high peak currents in short pulses without overheating, reduces overall power dissipation, and conserves battery life. Infrared LEDs are specifically optimized to operate with high-current, short-duration pulses, making this pulsed operation both safe and efficient.

From schematic:

$$R_4 = 47\Omega$$

$$V_{cc} = 9V$$

$$V_{FLED} = 1,3V$$

$$V_{ce(sat)} = 0,5V$$

$$Duty = 13\%$$

Peak current (I_{peak}):

$$I_{peak} = \frac{V_{cc} - V_F - V_{ce(sat)}}{R_4}$$

$$I_{peak} = \frac{9 - (2 \times 1,3) - 0,5}{47}$$

$$I_{peak} = 0,1255A$$

$$I_{peak} = 125mA$$

Average current (I_{avg}):

$$I_{avg} = I_{peak} \times Duty$$

$$I_{avg} = 0,1255 \times 0,13$$

$$I_{avg} = 0,017A$$

$$I_{avg} = 17mA$$



Role of the BD140 Transistor

The BD140 transistor provides the high-current switching stage in the circuit. It receives the low-current control signal from the 555 timer and switches the supply to the infrared LED string. Acting as a power switch, the transistor turns ON when the 555 output goes LOW, allowing current to flow through the LEDs, and turns OFF when the 555 output goes HIGH, cutting the current. This arrangement allows the transistor to:

- Control the LEDs efficiently
- Amplify the small control current from the 555
- Protect the 555 timer from excessive current draw

A transistor is necessary because the 555 timer cannot safely supply the peak current required by the infrared LEDs. The BD140 acts as a switch: when it is in saturation, it is fully ON and conducts current; when in cutoff, it is fully OFF and blocks current. The transistor's current gain (β or h_{FE}) determines how much base current is needed to drive the desired collector current. A higher β reduces the base current required from the 555.

In essence:

- The 555 acts as the control signal
- The BD140 acts as a current amplifier or power switch, providing the necessary high current to drive the LED string safely

Role of Capacitor C2

Capacitor C2 acts as a local energy reservoir that supplies current during fast LED pulses, preventing voltage drops in the supply. It stabilizes the transmitter and reduces interference by filtering switching noise, ensuring reliable operation.